GGG-C-412b
June 30, 1964

SUPERSEDING
Fed. Spec. GGG-C-412a
April 10, 1958

FEDERAL SPECIFICATION

CLAMP, MATERIAL LIFTING (PLATE, HORIZONTAL LIFT AND VERTICAL LIFT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers horizontal and vertical material-lifting clamps of the eccentric cam gripping types for use with various mobile and overhead cranes in lifting, moving or transporting flat steel plates and slabs commonly found in shipyards, supply depots, and repair facilities.

1.1.1 *Federal Specification coverage.* Federal Specifications do not cover all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, and styles.* Plate lifting clamps shall be of the following types, classes, and styles, as specified (see 6.2):

Type I—Horizontal.
- Class 1—Single cam, serrated grip.
- Class 2—Double cam, serrated grip.
- Class 3—Singe lever, smooth jaw.

Type II—Vertical.
- Class 1—Without cam grip safety lock.
- Class 2—With cam grip safety lock.
  - Style A—Without lock-open feature.
  - Style B—With lock-open feature.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

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Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Material lifting clamps furnished under this specification shall be products which have been tested, and passed the qualification tests specified herein, and have been listed on or approved for listing on the applicable qualified products list.

3.2 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of plate lifting clamps otherwise conforming to this specification.

3.3 Material and construction.

3.3.1 *Material.* Material lifting clamps shall be constructed of steel of a quality to produce clamps conforming to the requirements as hereinafter specified.

3.3.2 *Construction.* Material lifting clamps shall be of safe construction for their rated capacity and shall be capable of withstanding the respective tests specified in 4.4 through 4.5.4.2, as applicable. Clamps of a rated capacity greater than that specified in the applicable table will be acceptable provided they are marked with the minimum capacity specified in the applicable table and meet all other requirements of this specification. Clamps having a plate thickness capacity covering a wider range than that specified in the applicable table will be acceptable provided they are marked with the minimum range specified in the applicable table and meet all other requirements. The design shall be such that all parts may be readily removed and replaced. All parts shall be provided with adequate means to prevent accidental disengagement during normal operating conditions.

3.3.2.1 *Projections.* Projections on the external portions of the body or frame of clamps shall be held to a minimum. Pins, heads of rivets, and other portions of the body or frame shall be constructed in a manner that will minimize the possibility of catching or hooking on adjacent plates or objects.

3.3.2.2 *Cam grips.* The plate gripping surfaces of the cam grips shall have either a smooth machined pressure edge or serrations of suitable shape, depth, and surface area to assure a positive grip. The pressure edges or serrations and adjacent area shall be properly hardened and tempered for a distance extending at least 3/8 inch back from the working edges and shall show a minimum Rockwell C reading of 50 (see 4.5.2).

3.4 Repair parts. The contractor supplying clamps shall make available and furnish repair parts when requested. All parts having the same manufacturer's part number shall be constructed to definite standards, tolerances, and clearances to permit complete interchangeability, replacement, or adjustment without modification.

3.5 Marking. All material lifting clamps shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. Also the minimum rated capacity of each horizontal clamp (not the minimum rated capacity in pairs) and each verticle clamp as well as the minimum range of the plate thickness capacity in inches shall be metal die stamped, engraved, or embossed in plain view on each clamp. The minimum rated capacity shall be in tons or part thereof and shall be legible at a distance of not less than 10 feet. Marking

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shall be accomplished in such a manner that it will in no way interfere with the mechanism or satisfactory operation of the lifting clamp.

3.6 Finish. All surfaces shall be thoroughly cleaned and free from dirt, fins, scale, or other harmful or extraneous material. Unless otherwise specified in the contract or order (see 6.2), all nonbearing surfaces of the lifting clamps shall be finished with one coat of wash primer followed by one coat of enamel.

3.7 Type I, horizontal clamp. Type I clamps shall be designed for use in handling steel plates in a horizontal position. A serrated steel gripping pad inserted in the body or frame will be permitted, on classes 1 and 2 clamps and, if furnished, the serrations shall show a minimum Rockwell C hardness reading of 50 (see 4.5.2). A light blow shall cause release and allow disengagement of the clamp from a steel plate after lifting tension is released.

3.7.1 *Class* 1, *single cam grip.* **Type I, class** 1 clamps shall consist essentially of a body or frame, cam grip, and cam pin.

3.7.1.1 *Body or frame.* The body or frame shall be designed to receive the cam pin and allow the cam grip to clamp any plates within the plate thickness capacity.

3.7.1.2 The cam grip shall conform to 3.3.2.2 and shall be provided with a lifting eye (hole) to accommodate a suitable shackle or other attachment.

3.7.1.3 *Cam pin.* The cam pin may be hardened to increase its strength and minimize wear, but in such case it shall be properly drawn to eliminate brittleness.

3.7.1.4 Type I, class 1, clamps shall conform to table I, for the capacity specified (see 6.2), and shall be similar to A or B of figure 1.

TABLE I.—*Type I, class* 1, *horizontal clamp, single cam grip*

Rated capacity in pairs (min.)	Plate thickness capacity[1] (min. range)	Size of lifting eye (hole) (min.)	Weight (max.) in pairs
Tons	*Inches*	*Inches*	*Pounds*
2	0 to 1-1/2	1-3/8	35
6	0 to 2	1-3/8	100

[1] Does not preclude a greater range provided the minimum range specified is covered (see 3.3.2).

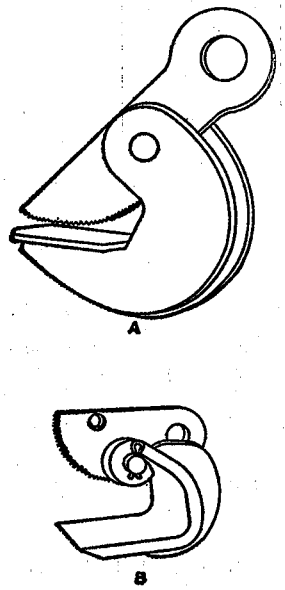


FIGURE 1.—Type I, class 1, single cam grip, horizontal clamp.

3.7.2 *Class* 2, *double cam grip.* Type I class 2 clamps shall consist essentially of a body or frame, cam grips, cam pin, and lifting shackle with pin.

3.7.2.1 *Body or frame.* The body or frame shall conform to 3.7.1.1

3.7.2.2 *Cam grips.* The cam grips shall conform to 3.3.2.2 and shall be provided with suitable holes to accommodate the lifting shackle pin.

3.7.2.3 *Cam pin.* The cam pin shall conform to 3.7.1.3.

3.7.2.4 *Lifting shackle with pin.* The lifting shackle and pin shall be of suitable de-

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sign and the minimum shackle opening shall be as specified in 3.7.2.5.

3.7.2.5 Type I, class 2 clamps shall conform to table II, for the capacity specified (see 6.2), and shall be similar to figure 2.

TABLE II.—*Type I, class 2, horizontal clamp, double cam grips*

Rated capacity in pairs (min.)	Plate thickness capacity[1] (min. range)	Size of shackle opening (min.)	Weight (max.) in pairs
Tons	*Inches*	*Inches*	*Pounds*
3	0 to 2	2	95
6	0 to 2	2	130

[1] Does not preclude a greater range provided the minimum range specified is covered (see 3.3.2).

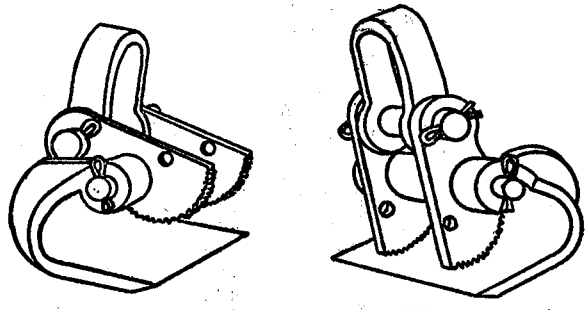


FIGURE 2.—Type I, class 2, double cam grip, horizontal clamp.

3.7.3 *Class 3, single lever, smooth jaw.* Type I, class 3 clamps shall consist essentially of a body or frame with a smooth wide jaw permanently affixed, a single lever fitted with a lifting ring, and a smooth upper jaw. The lever shall be attached to the frame by a pin, the upper jaw shall be attached to the lever by a swivel and shall assume a flat, positive grip on the material being lifted.

3.7.3.1 *Body or frame.* The body or frame shall be designed to receive a lever pin and allow the plate gripping jaws to clamp any plates within the thickness capacity.

3.7.3.2 *Lever pin.* The lever pin may be hardened to increase its strength and minimize wear but in such cases it shall be properly drawn to eliminate brittleness.

3.7.3.3 *Upper jaw pin.* The upper jaw pin shall conform to 3.7.3.2.

3.7.3.4 *Lifting ring.* The lifting ring shall be of a suitable design and shall be as specified in 3.7.3.5.

3.7.3.5 *Type 1, class 3.* Clamps shall conform to table III for the capacity specified (see 6.2) and shall be similar to figure 3.

TABLE III.—*Type I, class 3, horizontal clamp, single lever, smooth jaw grip*

Rated capacity in pairs (min.)	Plate thickness capacity[1] (min. range)	Size of lifting ring (opening) (min.)	Weight (max.) in pairs
Tons	*Inches*	*Inches*	*Pounds*
12	0 to 3	4	200

[1] Does not preclude a greater range provided the minimum range is covered (see 3.3.2).

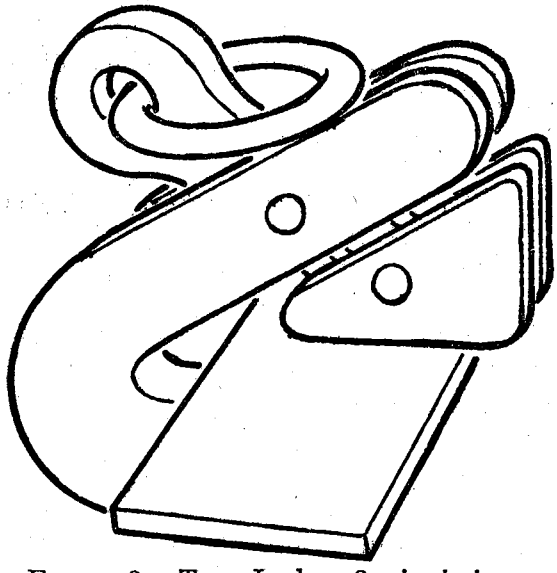


FIGURE 3.—Type I, class 3, single lever, smooth jaw, horizontal clamp.

3.8 **Type II, vertical clamp.** Type II clamps shall be designed for lifting and carrying steel plates vertically by gripping a plate resting in a horizontal or a vertical position. All parts shall operate freely and shall not bind under normal operating conditions.

3.8.1 *Class 1, without cam grip safety lock.* Type II, class 1 clamps shall consist es-

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sentially of a body or frame, lifting shackle, two cam grips or a single cam grip and gripping pad, and the connecting linkage. The clamp shall maintain its full grip on the steel plate as long as the lifting tension is applied by the crane. A light blow shall cause release and allow disengagement of the clamp from a steel plate after the lifting tension is released.

3.8.1.1 *Body or frame.* The body or frame of clamps having two eccentric cam grips shall have both jaws approximately equal in length. The body or frame of clamps having a single cam grip shall have a shorter jaw on the side opposite the cam grip to facilitate the entrance of steel plates between the jaws. The shorter jaw shall be provided with a gripping pad. All working parts with the exception of the lifting shackle shall be constructed within the body or frame.

3.8.1.2 *Lifting shackle.* The lifting shackle shall be constructed in such a manner that when a steel plate is lifted it will hang in practically a vertical position directly under the crane hook or sling attachment. The minimum shackle opening shall be as specified in 3.8.1.6.

3.8.1.3 *Cam grip(s).* The cam grip(s) shall conform to 3.3.2.2.

3.8.1.4 *Gripping pad.* The plate gripping surface of the gripping pad shall have serrations of suitable shape and depth to assure a positive grip. Suitable means may be provided to allow the pad to swivel (rotate) in any direction which contributes to alignment of the clamp with the direction of pull. The entire gripping pad or the serrations thereon shall be properly hardened to show a minimum Rockwell C reading of 50 (see 4.5.2).

3.8.1.5 *Connecting linkage.* All pins and connecting links shall be of suitable design to assure maximum service life under maximum service conditions.

3.8.1.6 Type II, class 1 clamps shall conform to table IV, for the capacity specified (see 6.2), and shall be similar to figure 4.

TABLE IV.—*Type II, class* 1, *vertical clamp, without cam grip safety lock*

Rated capacity (min.)	Plate thickness capacity[1] (min. range)	Size of eye or shackle opening (min.)	Weight (max.)
Tons	*Inches*	*Inches*	*Pounds*
1	0 to 3/4	1-7/8	22
3	0 to 1	2-1/2	30
5	1/4 to 1-1/2	3	60
10	1 to 2-1/2	3	115

[1] Does not preclude a greater range provided the minimum range specified is covered (see 3.3.2).

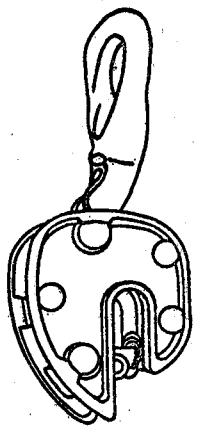


FIGURE 4.—Type II, class 1, vertical clamp, without cam grip safety lock.

3.8.2 *Type II, class* 2. With cam grip safety lock.

3.8.2.1 *Style A, without lock-open feature.* Type II, class 2, style A clamps shall consist essentially of a body or frame, cam grip safety lock, lifting shackle, cam grip gripping pad, and the connecting linkage.

3.8.2.1.1 *Body or frame.* The body or frame shall have jaws of unequal length, the shorter jaw being on the side opposite the cam grip in order to facilitate the entrance of steel plates between the jaws. The shorter jaw shall be recessed or other means provided for a positive support of the gripping pad. The actuating lever of the safety

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lock may be located within the body or frame, or mounted externally. All other working parts, with the exception of the lifting shackle, shall be constructed within the body or frame.

3.8.2.1.2 *Cam grip safety lock.* The safety lock mechanism shall be of such design that when the actuating lever is in the closed position, the cam grip shall be effectively locked onto a steel plate both with and without lifting tension being applied. When the actuating lever is moved to neutral or open position while a steel plate is suspended, release of the grip upon the plate shall not occur. The actuating lever shall be held in a neutral or open position in order to free the clamp after release of the lifting tension. Means shall be provided for attachment of a pull cord to the actuating lever. Any spring used in the mechanism shall be of high quality steel properly heat-treated to prevent breakage or permanent set under normal operating conditions.

3.8.2.1.3 *Lifting shackle.* The lifting shackle shall conform to 3.8.1.2, except the minimum shackle opening shall be as specified in 3.8.2.1.7.

3.8.2.1.4 *Cam grip.* The cam grip shall conform to 3.3.2.2.

3.8.2.1.5 *Gripping pad.* The gripping pad shall conform to 3.8.1.4.

3.8.2.1.6 *Connecting linkage.* All pins and connecting links shall be of suitable design to assure durability and maximum service.

3.8.2.1.7 Type II, class 2, style A clamps shall conform to table V for the capacity specified (see 6.2) and shall be similar to figure 5.

3.8.2.2 *Style* B, *with lock-open feature.* Type II, class 2, style B clamps shall conform to the requirements for class 2, style A clamps, except style B clamps shall be provided with a "lock-open" device. The "lock-open" device shall function as follows: Before a suspended steel plate is lowered to rest in an upright plate rack the actuating lever is moved to a neutral position by the rigger. When the steel plate is lowered to rest (causing a complete release of lifting tension) the actuating level automatically moves to the open position causing the "lock-open" device to lock the cam grip open (unable to lift). The clamp may then be raised from the steel plate without further action by the rigger.

3.8.2.2.1 Type II, class 2, style B clamps shall conform to table V for the capacity specified (see 6.2), and shall be similar to figure 5.

TABLE V.—*Type II, class 2, styles A and B, vertical clamp, with cam grip safety lock, with and without lock-open feature*

Rated capacity (min.)	Plate thickness capacity[1] (min. range)	Single eye or shackle opening (min.)	Weight (max.)
Tons	*Inches*	*Inches*	*Pounds*
1	0 to 3/4	1-7/8	23
2	0 to 1	3	30
4	0 to 1-1/4	3	46
8	1/2 to 1-1/2	3	74
12	1 to 2-1/2	3	142
20	1 to 3	3	266

[1] Does not preclude a greater range provided the minimum range specified is covered (see 3.3.2).

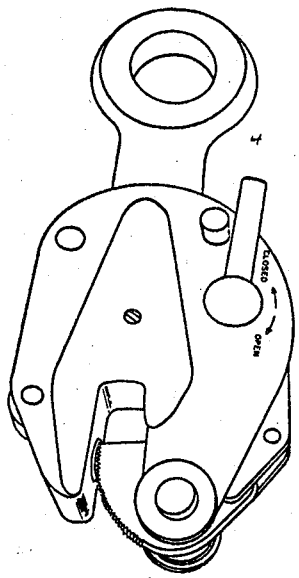


FIGURE 5.—Type II, class 2, style B vertical clamp with lock-open feature.

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3.9 Workmanship. Workmanship shall be first class in every respect. Edges and corners shall be rounded in conformance with good commercial standards. The clamps shall have no burrs, sharp projections, excessive surface roughness, or any other imperfections which may impair their durability and serviceability or create hazardous conditions under normal use.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 *Lot.* All material lifting clamps of the same type, class, style, and size presented at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 *Sampling for quality conformance.* A random sample of material lifting clamps shall be selected from each lot in accordance with MIL-STD-105 at inspection level II.

4.3 Classification of inspection. Testing of material lifting clamps shall be classified as follows:

(*a*) Qualification testing.
(*b*) Quality conformance inspection.

4.3.1 *Qualification testing.* Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification" see (6.3 and 6.4). The manufacturer shall submit a copy of his drawings for each different type, class, style, and size clamp submitted for qualification testing. These drawings will become a part of the laboratory report submitted to the Bureau of Ships. Manufacturers shall submit 4 clamps of each size, type I, class 1, 2 or 3 for which he desires qualification and 2 clamps of each size, type II, class 1, class 2, type A or class 2, type B he desires qualified. Qualification testing shall consist of the examination specified in 4.4 and the following tests. Type I, classes 1, 2 and 3, two each of each size shall be subjected to the tests specified in 4.5.4 and 4.5.4.1 and the remaining 2 each to the tests specified in 4.5.1.1, 4.5.2 and 4.5.3. Type II, class 1, class 2, type A, and class 2, type B, 1 each of each size shall be subjected to the tests specified in 4.5.4 and 4.5.4.1 and the remaining 1 each of type II, class 1 shall be tested as specified in 4.5.1.2, 4.5.2 and 4.5.3. The remaining 1 each of type II, class 2, type A, and class 2, type B shall be tested as specified in 4.5.1.3, 4.5.2 and 4.5.3.

4.3.2 *Quality conformance inspection.* Quality conformance inspection shall be applied to each sample selected in accordance with 4.2.2. Failure to pass the examination of one or more of the tests specified shall be cause for rejection. Quality conformance inspection shall consist of the following:

(*a*) Examination (4.4).
(*b*) Tests (4.5.1, as applicable, and 4.5.2).
(*c*) Inspection of preservation, packaging, packing, and marking for shipment and storage.

4.4 Examination. Each of the sample material lifting clamps selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Examination shall be conducted as specified in table VI with the acceptable quality level (AQL) as indicated. Any material lifting clamp in the sample containing one or more defects shall not be offered for delivery. If the number of defective material lifting clamps in

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any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

TABLE VI.—*Classification of defects in accordance with MIL-STD-105*

Categories	Defects
Critical:	None defined.
Major:	*AQL* 1.5 *percent defective.*
101	Clamp not type, class, or style specified.
102	Evidence of use of unauthorized material.
103	Rated capacity (in pairs) not as specified.
104	Minimum plate thickness capacity not as specified.
105	Size of lifting ring (opening) inadequate.
106	Working depth of throat (minimum) not as specified for nominal size.
107	Weight of clamps (in pairs) not as specified.
108	Cam grips do not have specified pressure edges or suitable shapes.
109	Body or frame not designed to allow cam grip to clamp all plates within rated plate capacity.
110	Jaw not smooth (type I, class 3 only).
111	Cam grip safety lock inadequate (type II, class 2 only).
112	Lock-open feature unsuitable or unsafe, (type II, class 2, style B only).
Minor:	*AQL* 2.5 *percent defective.*
201	Clamps not permanently marked; markings not legible; Source of manufacture not indicated.
202	Clamps not free of burrs, fins, scale, sharp, rough edges; not smooth finished.
203	Clamps not preserved or cleaned as specified.
204	Packing and packaging not as required.

4.5 Test procedures.

4.5.1 *Proof tests.*

4.5.1.1 *Type* 1, *classes* 1, 2 *and* 3, *horizontal clamps.* Type 1 clamps shall be tested in pairs with plates of suitable thickness depending upon jaw opening of clamps under test. Each pair of clamps shall be subjected to a load equal to three times the rated capacity of the pair, while mounted in the normal operating position, for a period of 2 minutes, either by the attachment of weights to the plate, or they may be tested singly or in pairs in a suitable testing machine. Subsequent to the load test, each clamp shall be visually examined for signs of damage or distortion and shall be examined to determine whether the clamp operates satisfactorily without binding of any part. Any clamp showing damage, distortion, or malfunctioning shall be rejected.

4.5.1.2 *Type II, class* 1, *vertical clamp.* Type II, class 1 clamps shall be tested with plates of suitable thickness depending upon jaw opening of clamps under test. Each clamp shall be subjected to a load equal to three times the rated capacity of the clamp, for a period of 2 minutes, either by the attachment of weights to the plate or in a suitable testing machine, while mounted in the normal operating position. Subsequent to the load test, each clamp shall be visually examined for signs of damage or distortion and shall be examined to determine whether the clamp operates satisfactorily without binding of any part. Any clamp showing damage, distortion, or malfunctioning shall be rejected.

4.5.1.3 *Type II, class* 2, *vertical clamp.* Test of type II, class 2 clamps shall be conducted under the same conditions as specified for type II, class 1, except that the safety cam lock mechanism shall be examined, before and after the load test, for signs of damage, distortion or malfunctioning. If the safety cam lock mechanism of any clamp shows damage, distortion, or malfunctioning before or after the load test, the clamp shall be rejected.

4.5.2 *Hardness test.* The cam grip(s) and gripping pad on each of the applicable clamps shall be tested to determine conformance to 3.3.2.2, 3.7, and 3.8.1.4, as applicable. Preparation for hardness tests shall be accomplished in such manner that will not impair the part for service. Measurements shall be made as close to the work-

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ing surface as practicable. Therefore, only the necessary minimum amount of scale should be removed at the point of measurement. Any cam grip or tripping pad which does not conform to the hardness specified shall cause rejection of the clamp.

4.5.3 *Breaking test.* The clamp under test shall grip a suitable test specimen having a minimum thickness of 75 percent of the maximum thickness range specified for the applicable clamp. The clamp shall be loaded in a suitable testing machine and the load increased until breakage occurs. The breaking load for each clamp shall be equal to, or greater than five times the rated capacity of the clamp. Any clamp with a breaking load less than five times the rated capacity of the clamp shall be rejected.

4.5.4 *Practical service test* .Type I clamps shall be tested in pairs and shall be required to lift steel plates from a horizontal position only. The plate shall be lifted, then landed and lifted again to test the clamps' ability to grab securely from a fully released position. The plate shall then be carried by normal crane movement along the track for a distance of approximately 50 feet and returned, testing holding qualities of clamps while crane is in motion. Type II clamps shall be tested singly and shall lift steel plates from both horizontal and vertical positions. The clamp shall engage and lift a plate from a horizontal position to a vertical position above the floor. The plate shall then be lowered to the floor in a vertical position and a light slack created by lowering the crane hook, the clamp shall grab as the plate falls thus testing the grabbing qualities. The plate shall then be carried by normal crane movement along the track for a distance of approximately 50 feet and returned, testing holding qualities of clamps while crane is in motion. The plate shall then be lowered into an upright plate rack and the clamp shall be easily released and removed when the clamp supporting rope has become slack and a light blow is applied to class 1 clamps and the safety lock actuating lever is in the open position for class 2 clamps.

4.5.4.1 *Off-balance test.* A steel plate that is approximately three times as long as its width and having a weight within 15 percent of the maximum rated capacity of the clamp(s) shall be used for this test. The clamp(s) shall engage the plate at such point that the distance from one end of the plate to the center of the clamp(s) shall be between 2 and 3 percent greater than one-half the over-all length of the plate. The plate shall then be lifted clear of the floor and allowed to shift into balance, testing the ability of the clamp(s) to hold during the shift.

4.5.4.2 *Inspection during testing.* During and after completion of the practical service test and the off-balance test, evidence of binding or jamming of working parts that are designed to move freely, or damage to the gripping serrations or any other part of the clamp shall be cause for rejection. Close observation shall also be made to determine if the clamps satisfactorily perform such functions as the following:

(*a*) The ability of the clamp to grip and hold securely various thicknesses of plates within the capacity specified in the applicable table.

(*b*) Easy engagement and disengagement of plates.

(*c*) Effective functioning of the safety lock mechanism on type II, class 2 clamps after repeated operations and without requiring adjustment.

4.6 Inspection of preparation for delivery. Sample clamps, packages, and packs shall be selected and inspected to verify conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of preservation, packaging, and packing).

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5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-H-15424. Levels of preservation and packaging and level of packing shall be as specified (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military activities.* Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. All clamps covered in this specification are intended to be used with cranes for lifting and moving flat steel plates and slabs.

6.1.1 Type I, classes 1, 2, and 3 clamps should be used in pairs (two or four clamps) with suitable slings and attachments.

6.1.2 Type II, class 1, and type II, class 2, styles A and B clamps may be used singly. However, safe practice regulations at some activities require the use of two vertical clamps with a sling and attachments in making each lift. The activating lever on style B clamps must be in a closed position when lifting steel plates from a horizontal to a vertical position.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type, class, and style (if applicable) of clamps required (see 1.2.1).
(*c*) Finish of nonbearing surfaces if different from 3.6.
(*d*) Rated capacity in pairs of type I clamps required (see 3.7.1.4, 3.7.2.5, and 3.7.3.5).
(*e*) Rated capacity of each for type II clamps required (see 3.8.1.6, 3.8.2.1.7, and 3.8.2.2.1).
(*f*) Level of preservation, packaging, and packing required (see 5.1).
(*g*) Special marking, when required (see 5.2.1).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL—GGG-C-412, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington, D. C. 20360 and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.4).

6.4 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

6.5 When ordering repair parts, specify the following:

(*a*) The manufacturer's name or trademark of the clamp for which the parts are required.
(*b*) Type, class, style (if applicable) and capacity of the clamp.
(*c*) Quantity of each part required.

6.6 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Clamps, cable, iron or steel.
Carload minimum weight 40,000 pounds.

Motor:

Clamps, cable, iron or steel.
Truckload minimum weight 40,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Army—MO

Navy—Ships

Air Force—WRA

Review:

Army—MO

Navy—Ships

Air Force—84

Preparing activity:

Navy—Ships

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